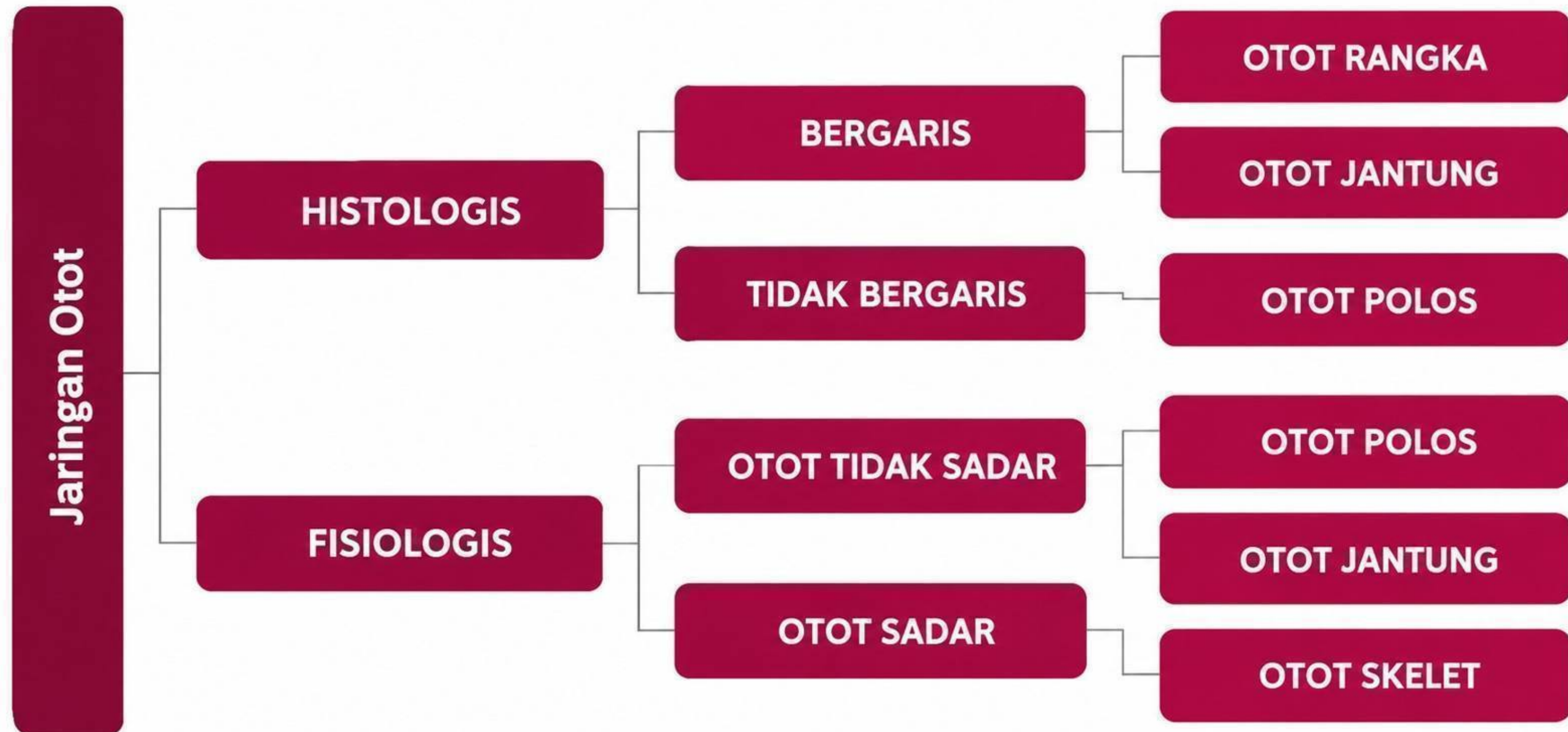




MUSCLE TISSUE

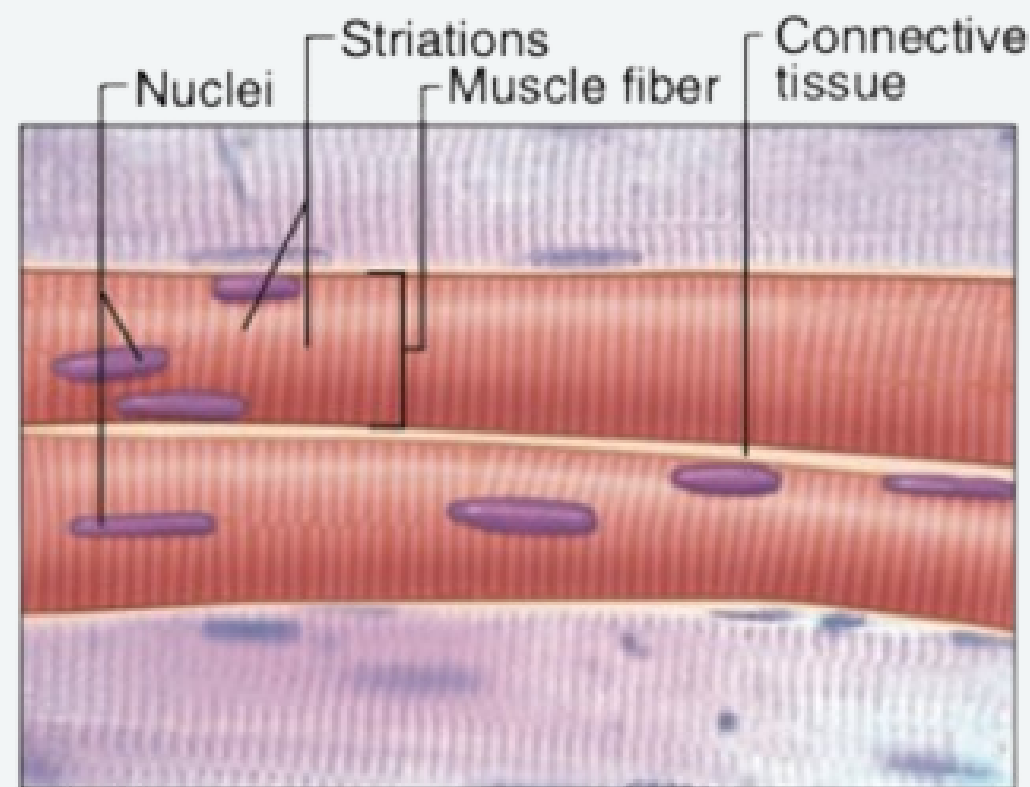
A tissue composed of a collection of muscle cells characterized by a large number of cytoplasmic filaments that are contractile.



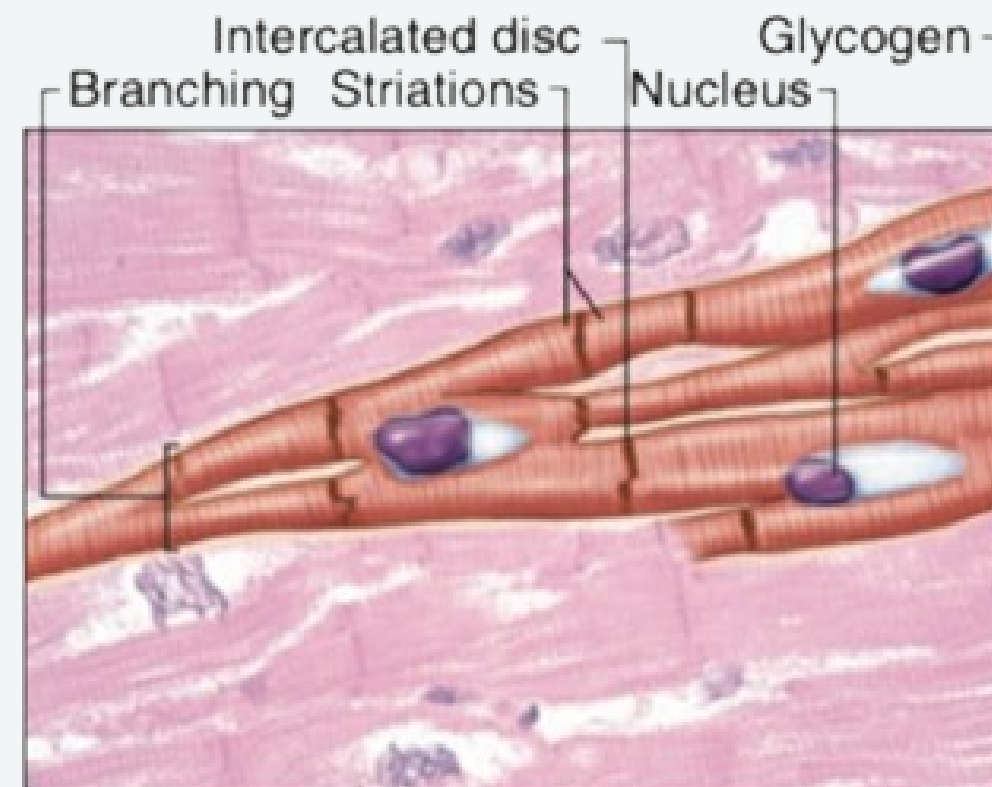




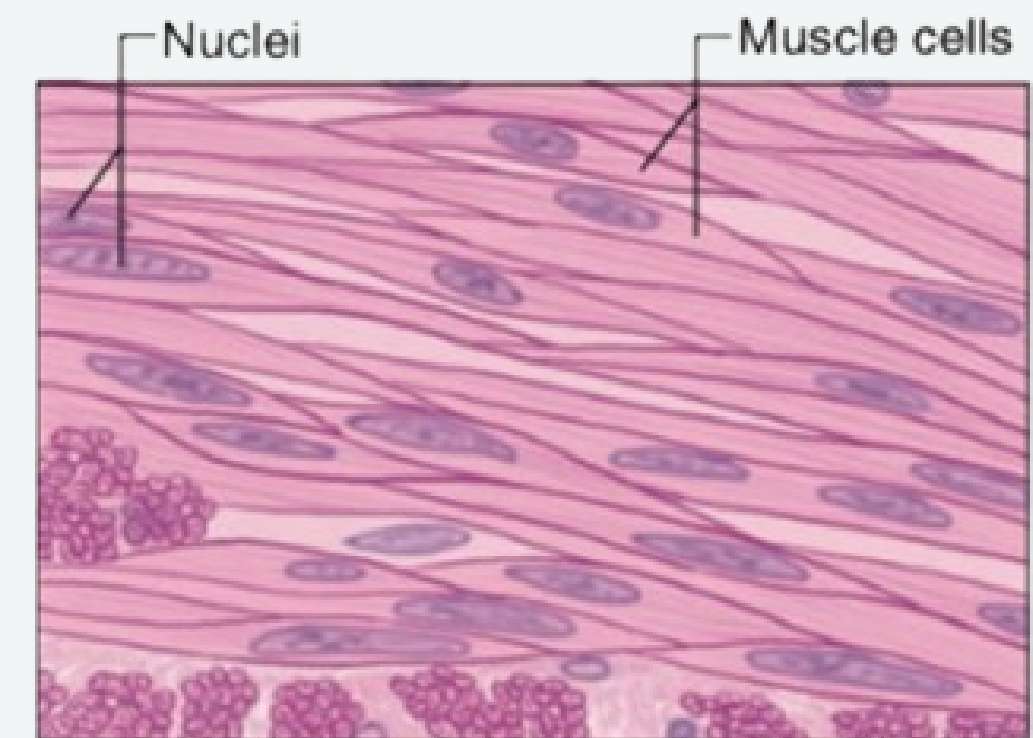
Types of Muscle



a Skeletal muscle



b Cardiac muscle



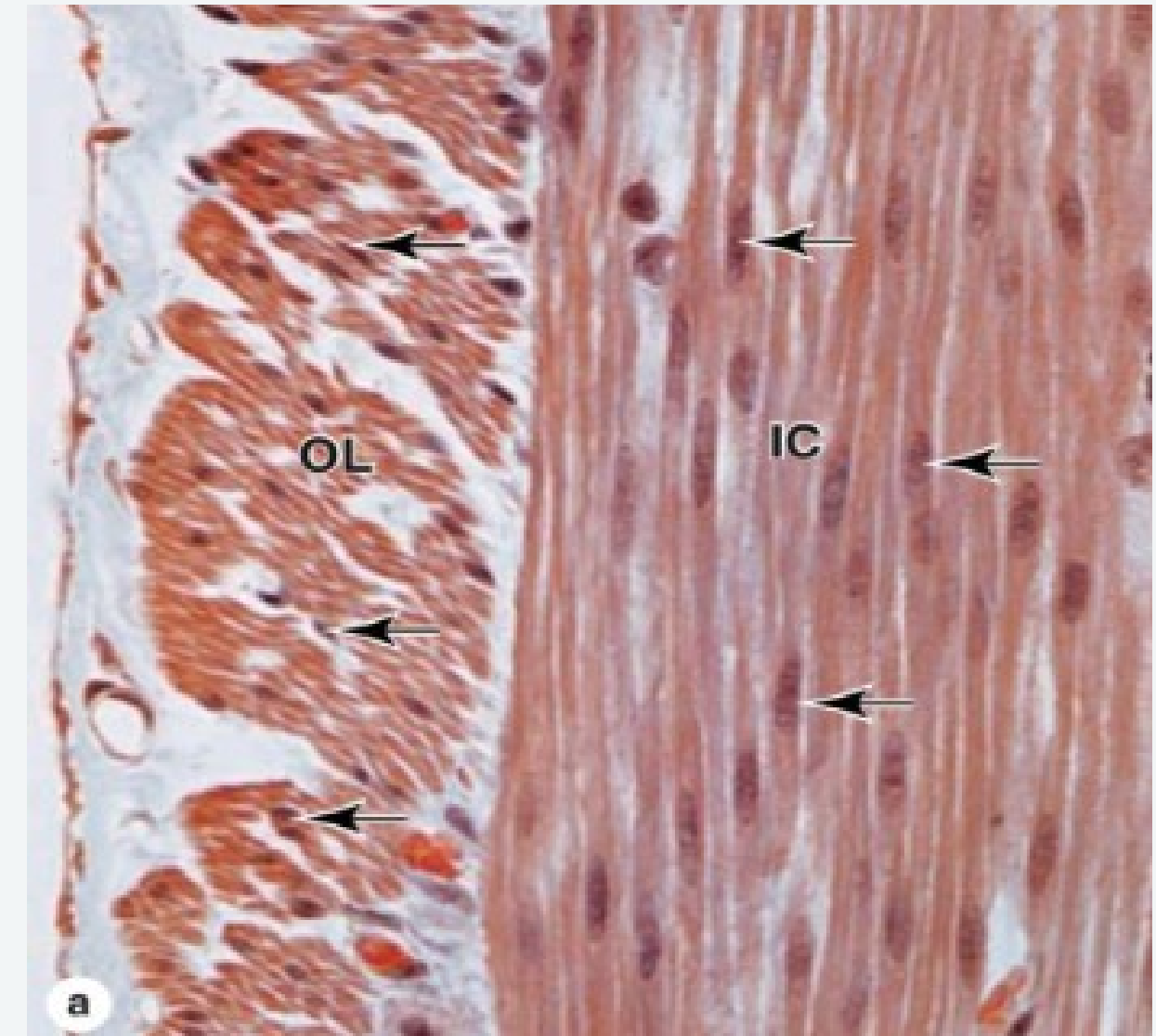
c Smooth muscle





Types of Histological Sections

- Membujur - Longitudinal - IC
- Melintang - Transversal - OL





Important Terms

- Sarcoplasm = cytoplasm of a muscle cell
- Sarcoplasmic reticulum = endoplasmic reticulum of a muscle cell
- Sarcosome = mitochondria
- Sarcolemma / plasmalemma = muscle cell membrane
- Myofiber = muscle cell
- Myofibril = subunit of a muscle cell





SMOOTH

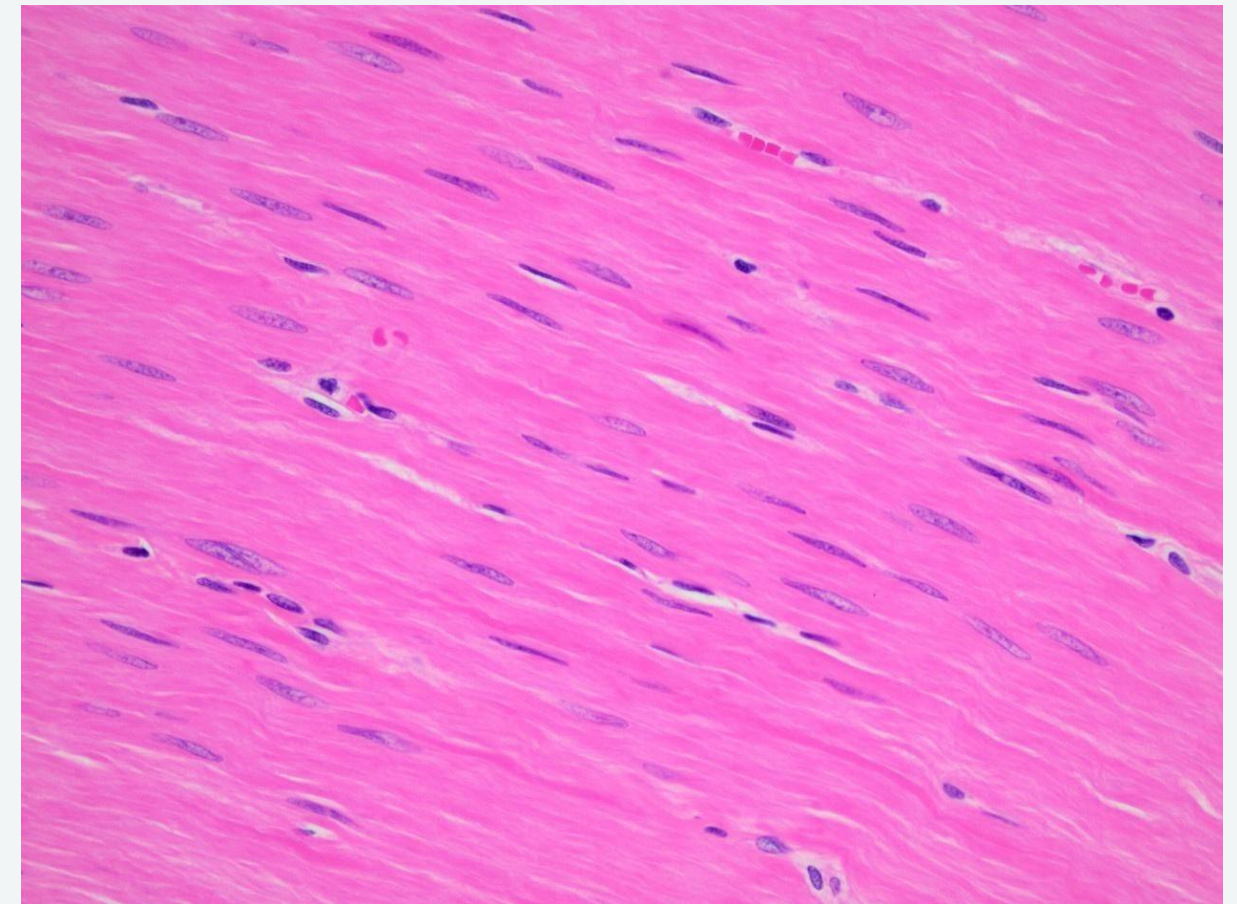
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Smooth Muscle

- Consists of fusiform cells without striations, characterized by slow and involuntary contractions; also known as visceral muscle.
- Surrounded by endomysium.
- Locations: blood vessels, gastrointestinal tract, respiratory tract, uterus, urinary bladder, and other internal organs.
- Innervation: autonomic nervous system. The myoneural junction is referred to as a napse.

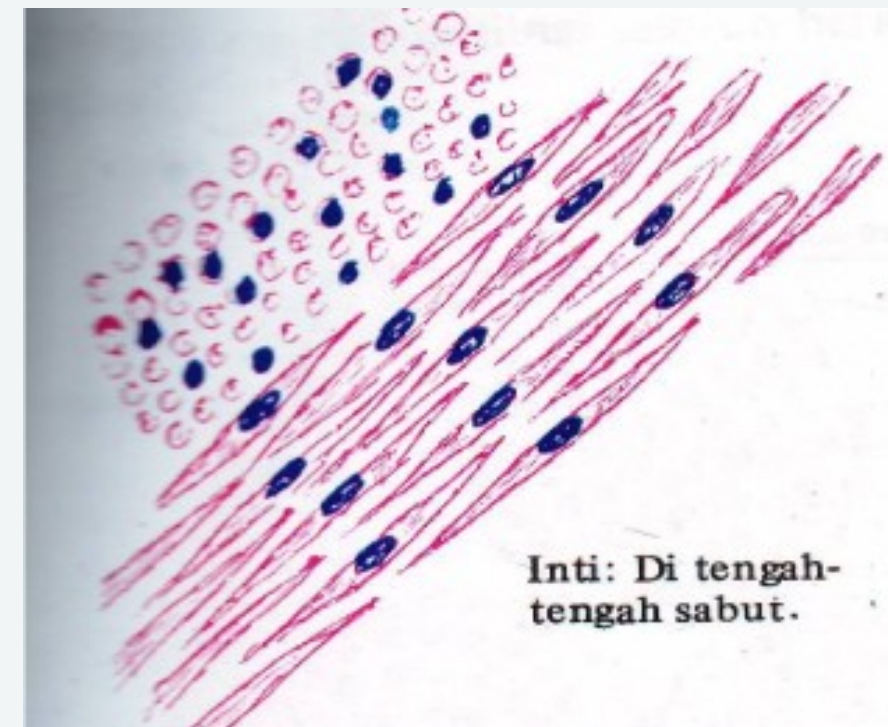




Smooth Muscle

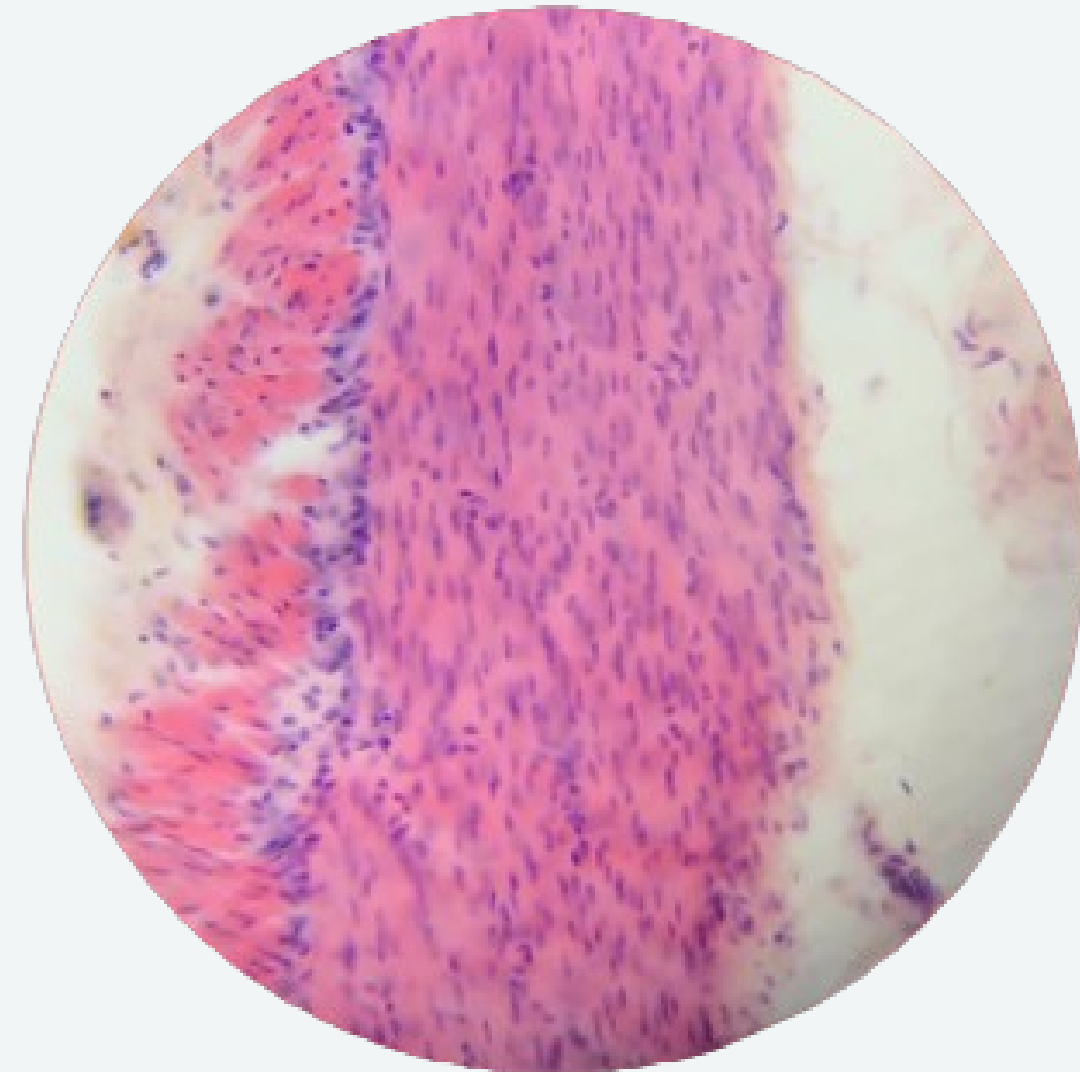
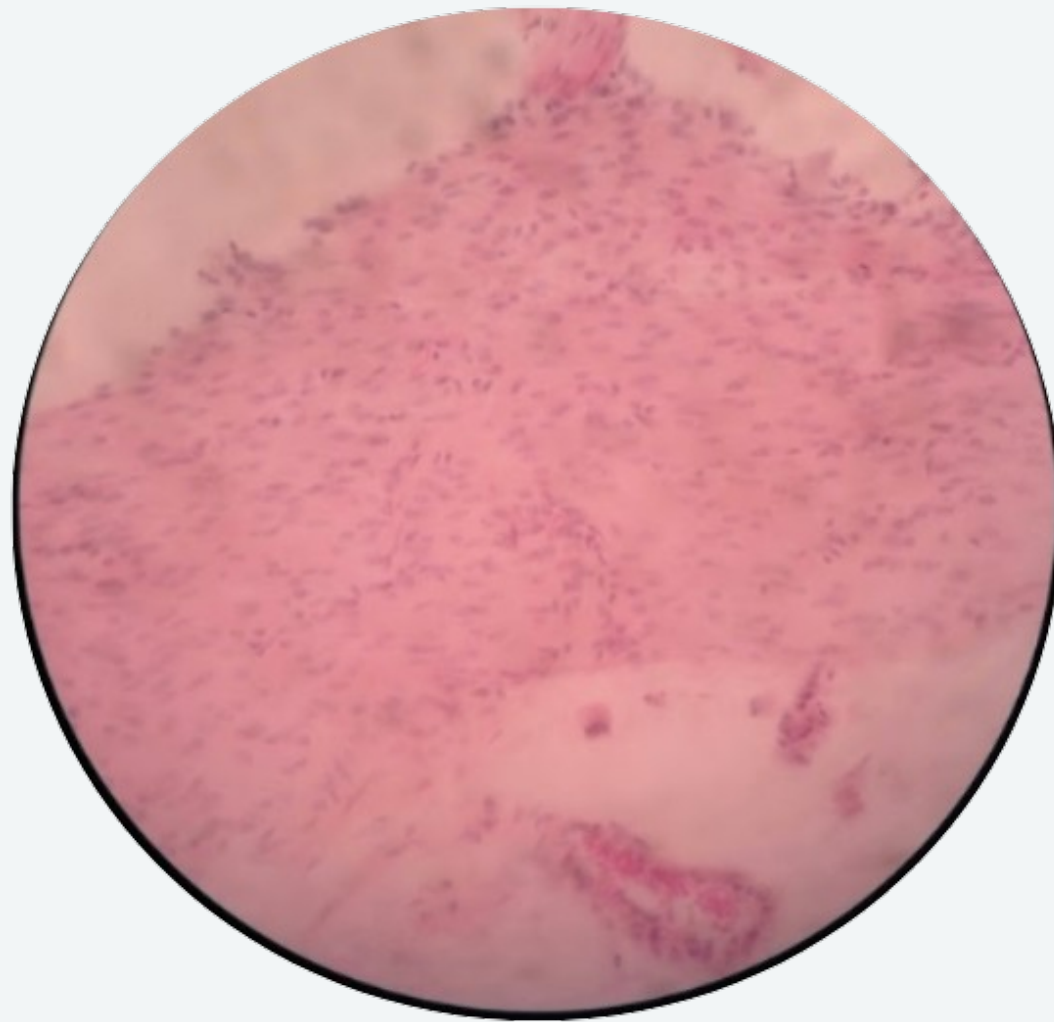
Histological Features:

- Fusiform-shaped cells
- No visible cross-striations
- Elongated, centrally located nucleus





Microscopic View





SKELETAL

MUSCLE

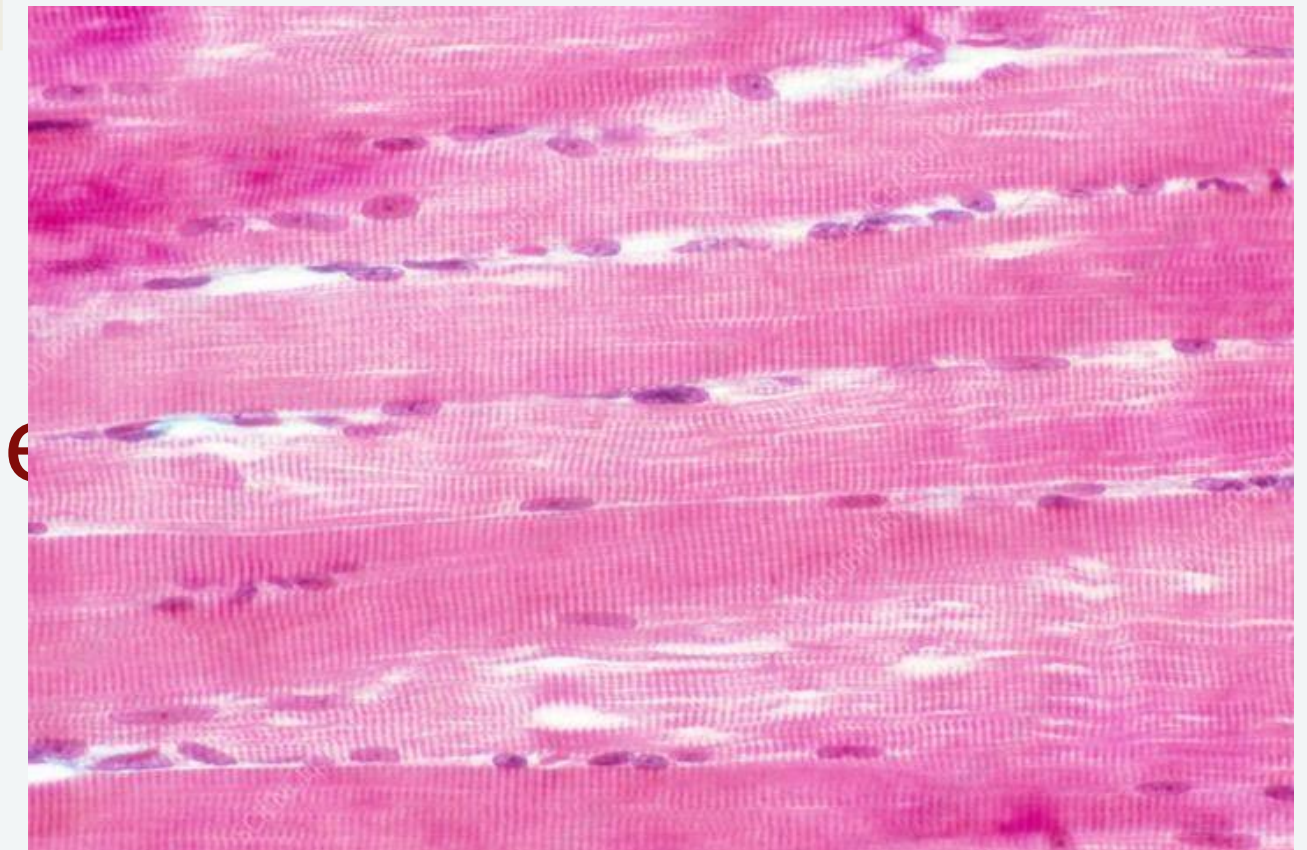
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Skeletal Muscle

- Also known as skeletal muscle / striated muscle / cross-striated muscle.
- Contracts voluntarily.
- Contraction is regulated by acetylcholine (ACh) released at the motor end plate (MEP) or neuromuscular junction (NMJ).
- Location: muscles of the limbs (extremities).

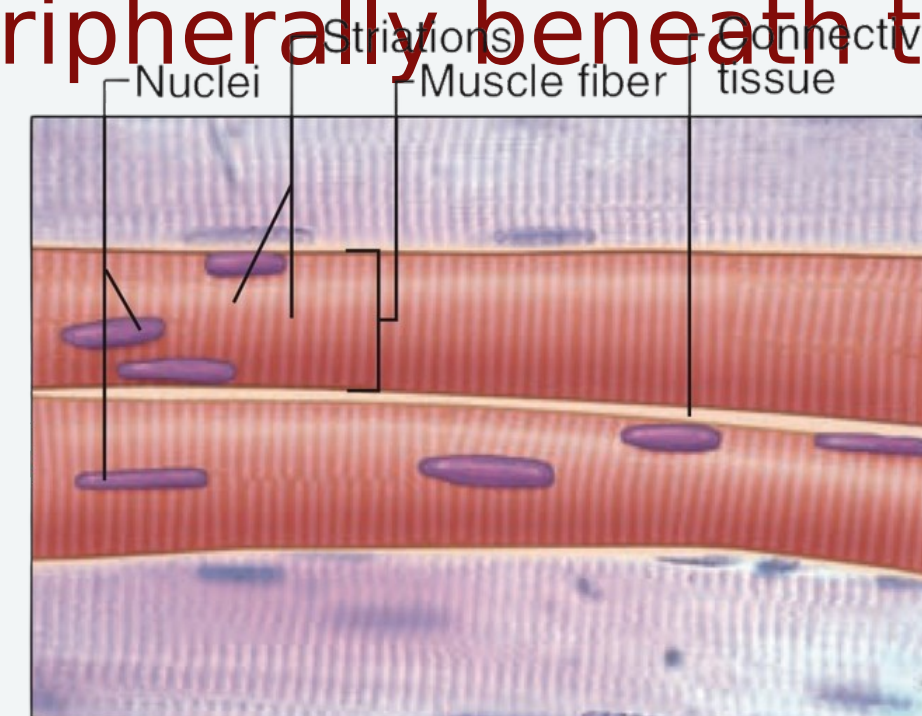
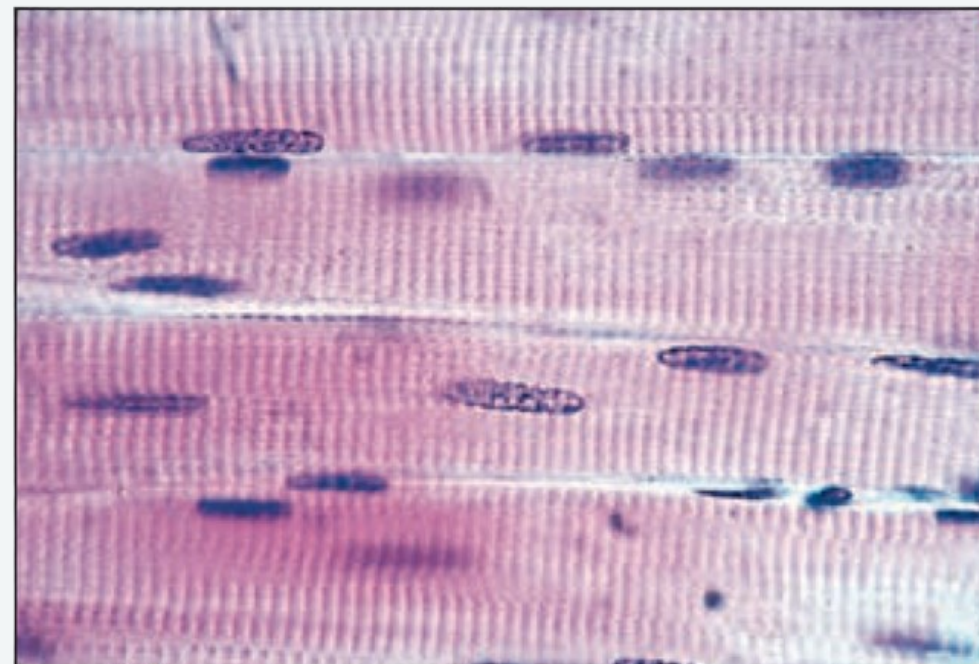




Skeletal Muscle

Histological Features

- Cylindrical in shape with multiple nuclei (more than one nucleus).
- Sarcolemma is clearly visible.
- Nuclei are oval-shaped and located peripherally beneath the sarcolemma.



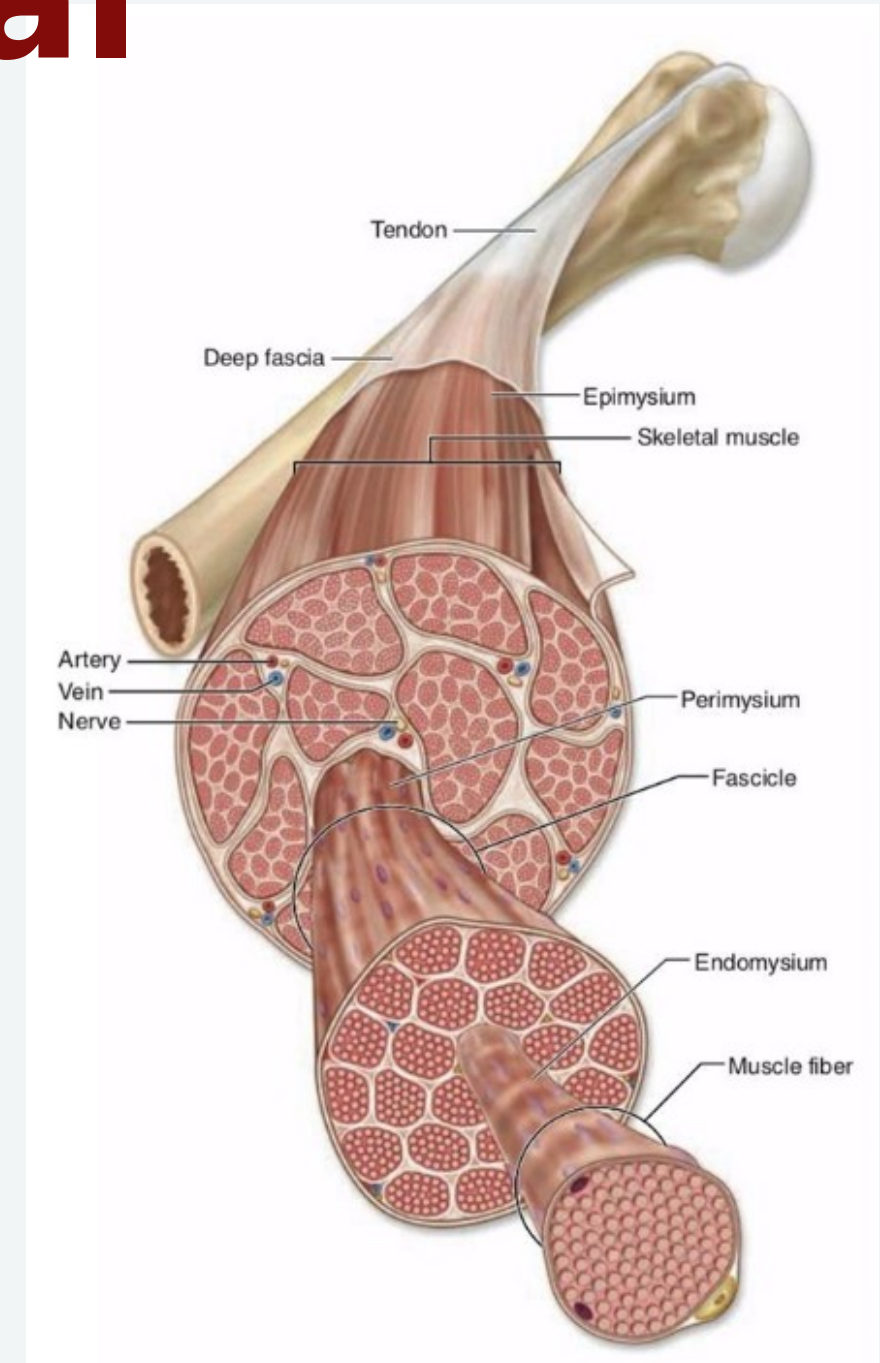
(a) Skeletal muscle





Structure of Skeletal

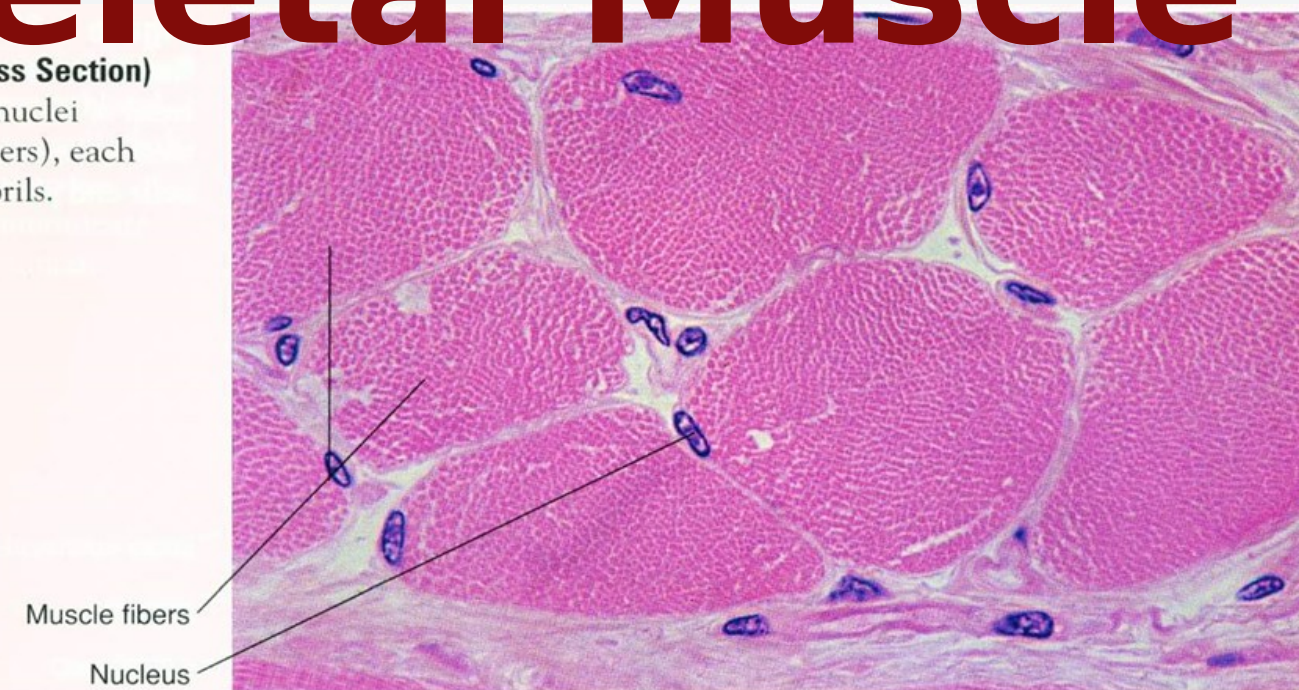
- **Epimysium:** the outermost layer of dense irregular connective tissue that surrounds the entire muscle.
- **Perimysium:** a layer of dense irregular connective tissue that surrounds a muscle fascicle.
- **Endomysium:** a very thin layer of fine reticular fibers surrounding each muscle fiber/myofiber.
- **Cross-section of a myofiber:** bundles of myofibrils are arranged within the sarcoplasm and appear as small islands called Cohnheim's areas.





Skeletal Muscle

Figure 1-38
Striated (Skeletal) Muscle (Cross Section)
Eccentrically located multiple nuclei accompany individual cells (fibers), each of which contains many myofibrils.
Human tongue. ($\times 250$)



Transversal section:

- Epimysium, perimysium, fascicles, and muscle fibers surrounded by endomysium are visible.
- Bundles of myofibrils appear in cross-section as island-like structures called Cohnheim's areas.
- Nuclei are located peripherally.





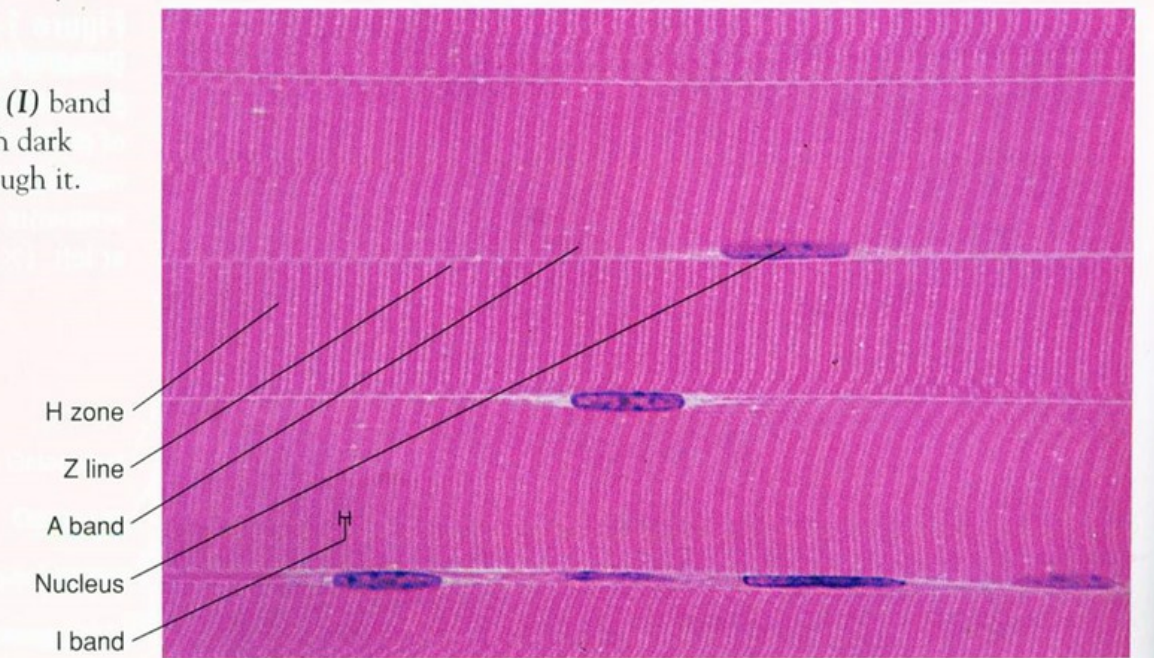
Skeletal Muscle

Longitudinal Section:

- Alternating light and dark bands are visible.
- Dark band → A band (anisotropic)
- Light band → I band (isotropic)
- At the center of the I band, there is a dark transverse line called the Z line (Z disc).
- At the center of the A band, there is a dark transverse line called the M line.
- At the center of the A band, there is a lighter transverse region called the H zone.

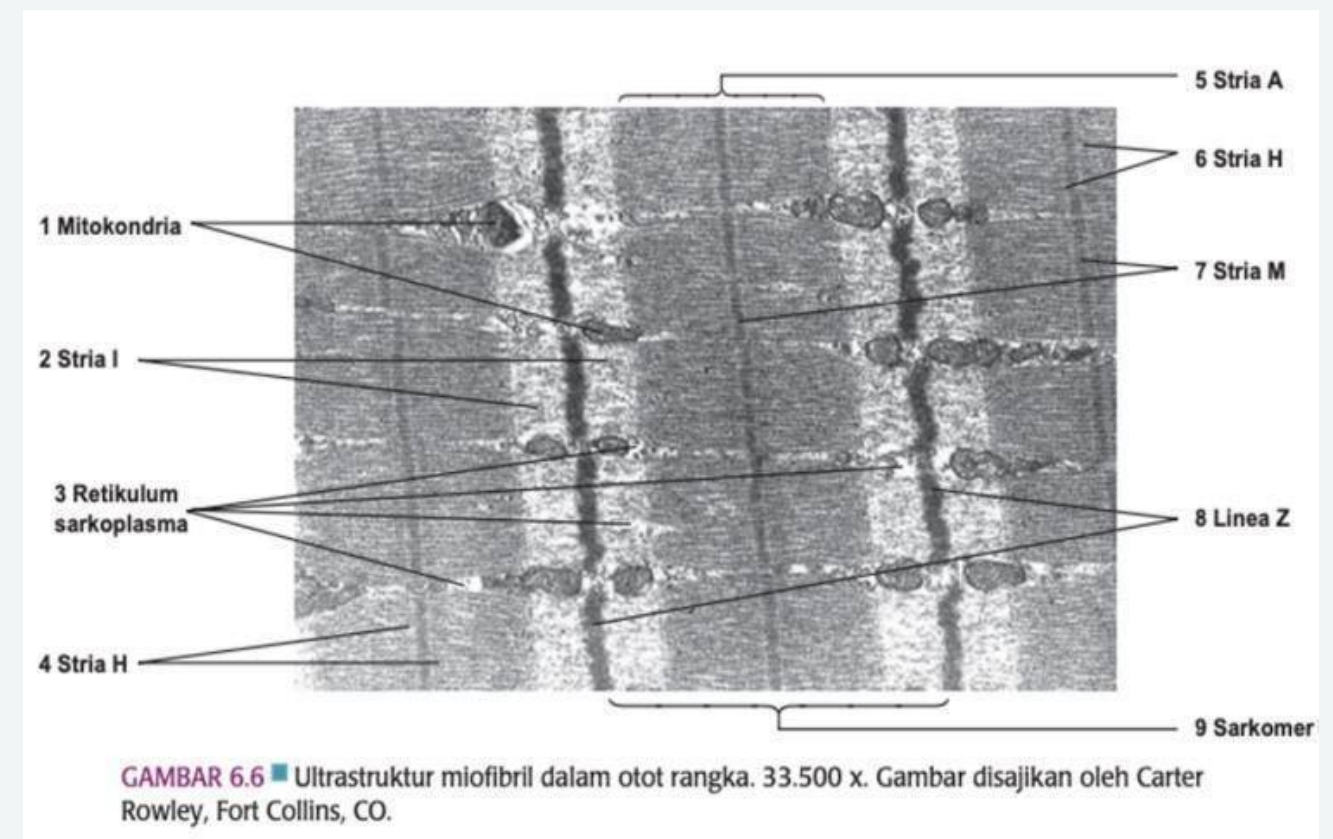
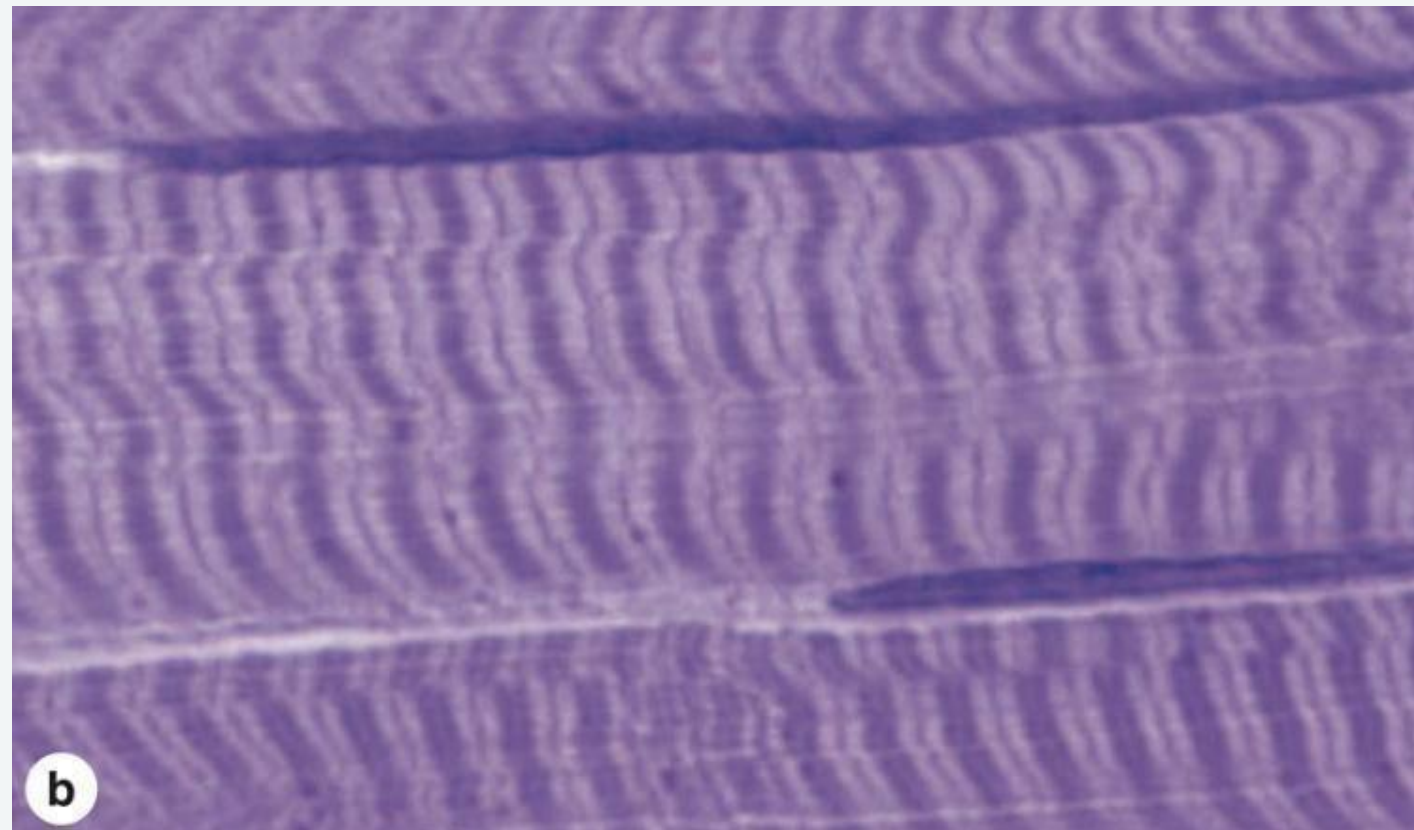
Figure 1-40

Striated (Skeletal) Muscle Fibers (Longitudinal Section) Each light (I) band has a dark (Z) line through it. Each dark (A) band has a light (H) zone through it. (×250)



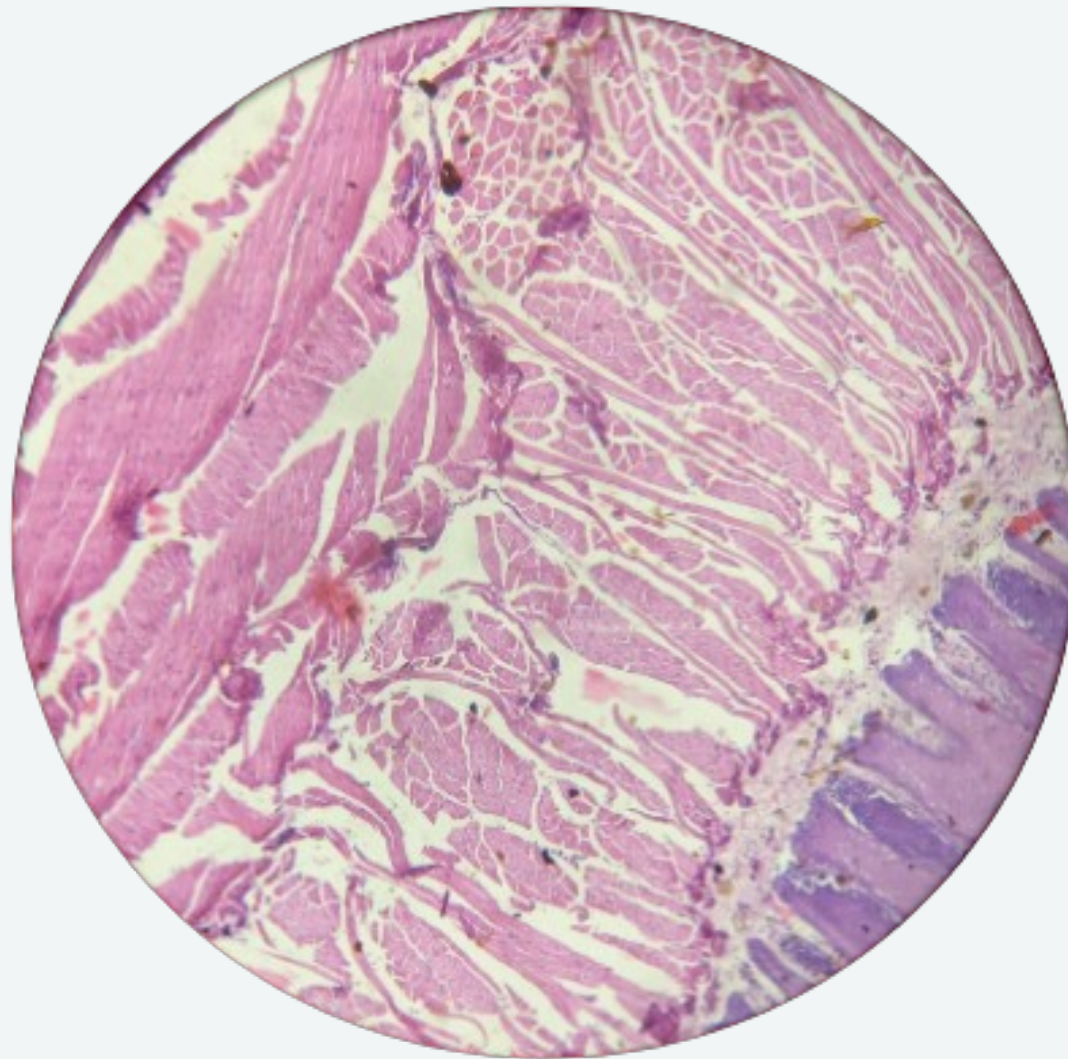


Skeletal Muscle





Microscopic View



**Transverse & Longitudinal
Sections**

Muscle bundles are visible



Transverse Section

Muscle sheath and Cohnheim's
areas are visible.



Longitudinal Section

Muscle bundles are visible.





CARDIAC

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Cardiac Muscle

Histological Features:

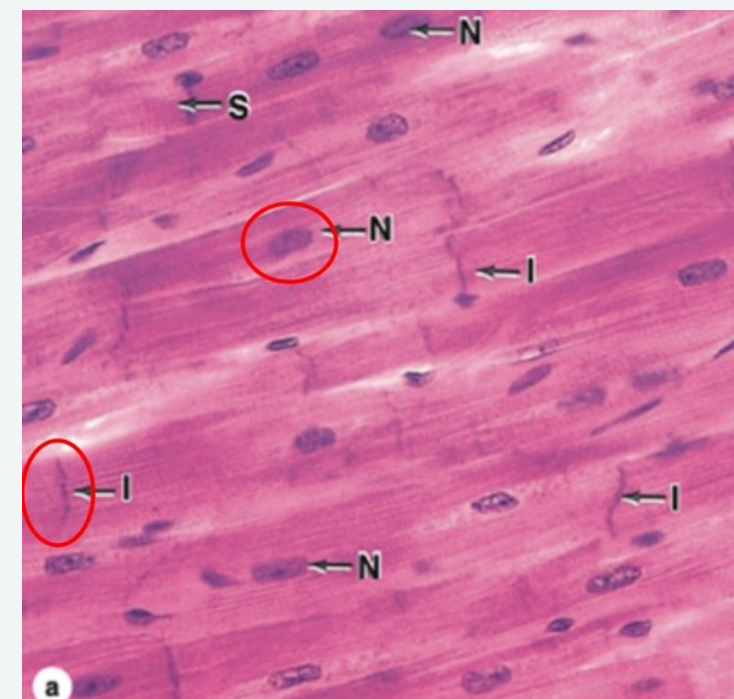
- Muscle fibers run parallel and are striated.
- Cardiac muscle fibers form intercellular anastomoses with one another.
- Irregular dark transverse lines called intercalated discs are visible.
- The nucleus is large, oval-shaped, and centrally located. It is usually single, but may occasionally be double.



Figure 1-41
Cardiac Muscle (Longitudinal Section)
Multinucleated, striated muscle fibers branch and anastomose at junctions marked by dark intercalated disks. ($\times 250$)

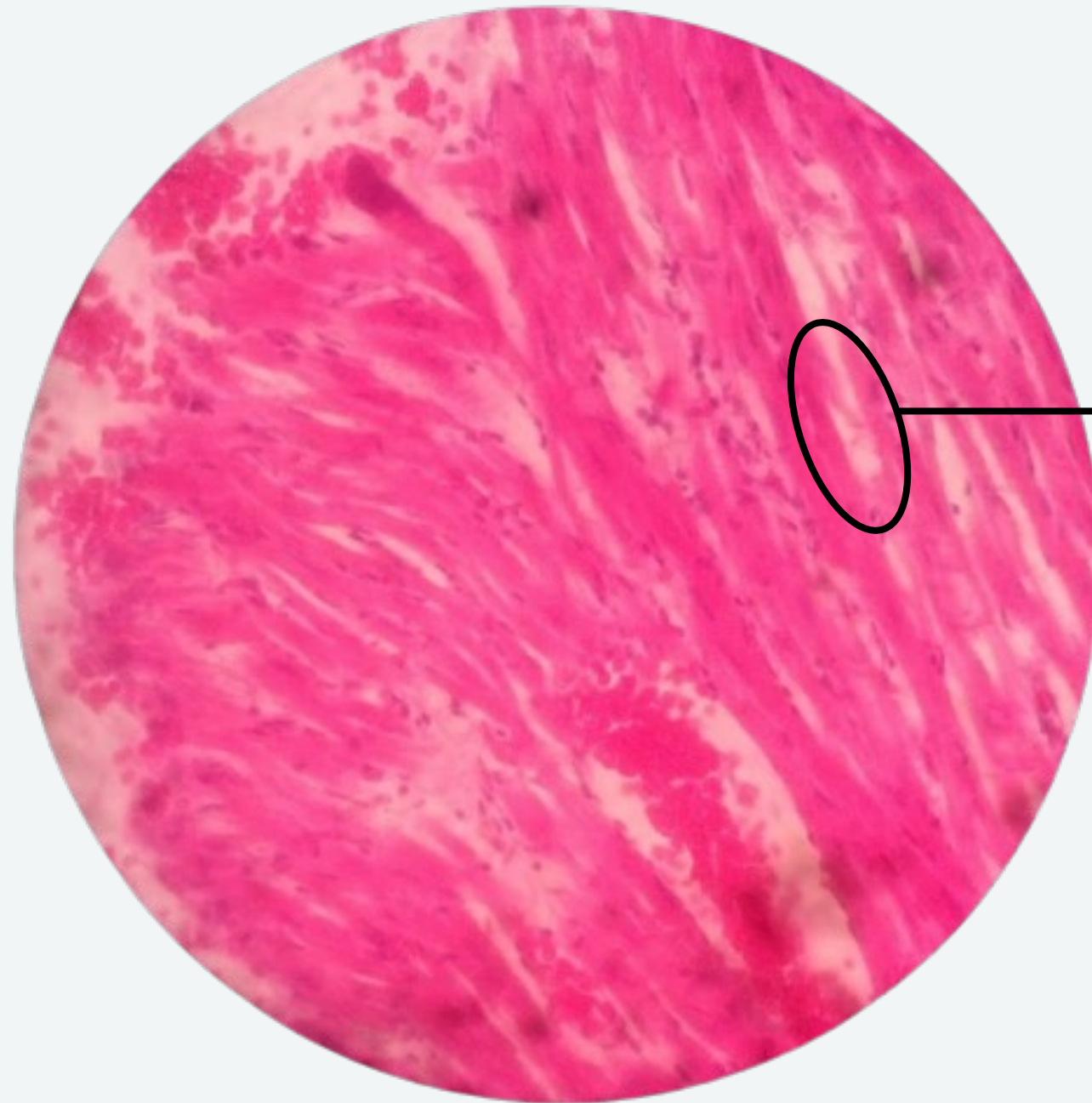
Nucleus

Intercalated disk





Cardiac Muscle

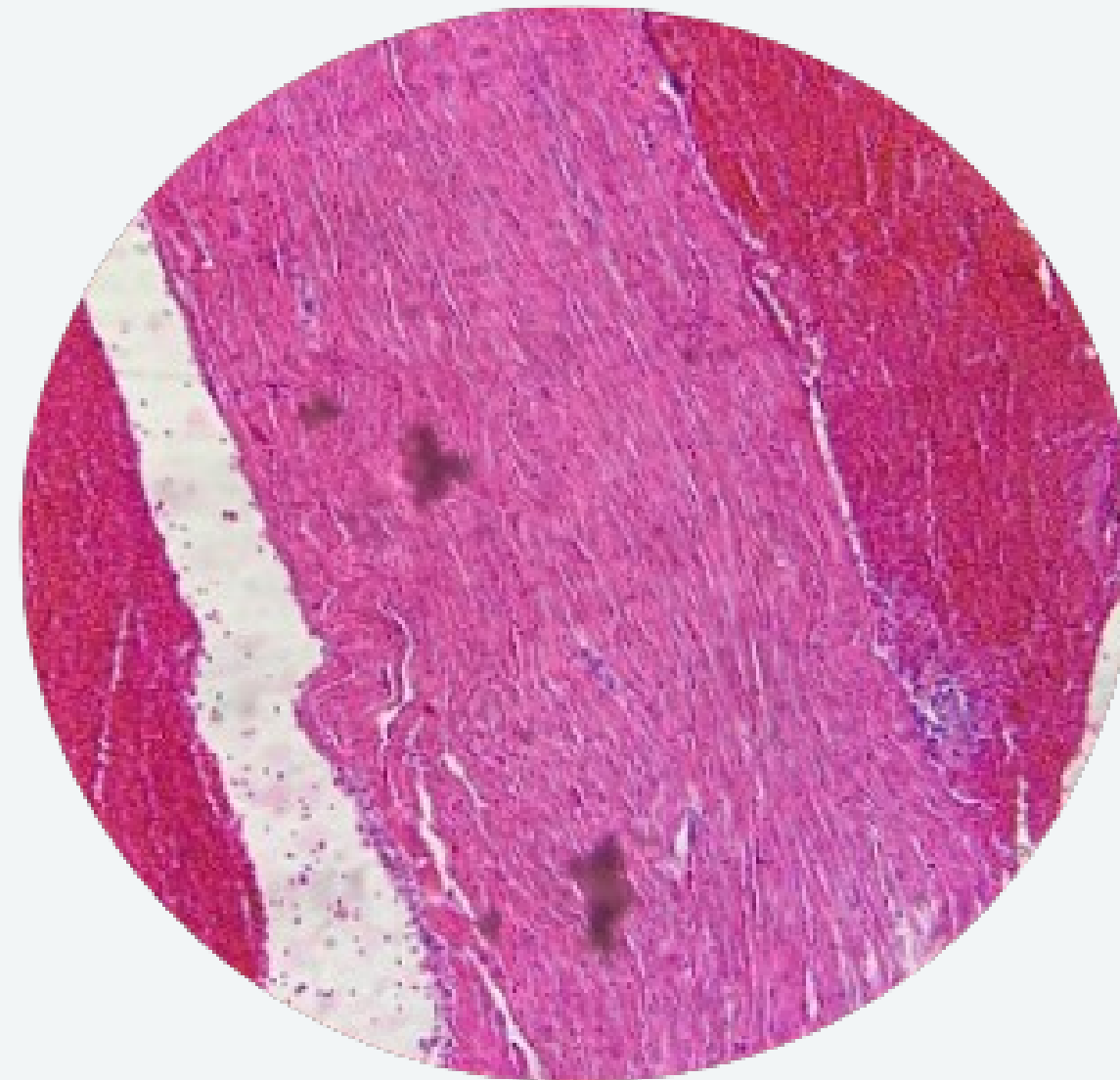


**Anastomose
serabut
otot**





Cardiac Muscle





SUMMAR

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	Otot Polos	Otot Jantung	Otot Lurik
Bentuk	Fusiform	Silindris Bercabang	Silindris
Letak Inti	Tengah	Tengah	Tepi
Pembelahan	Secara mitosis	Tidak ada tanda regenerasi setelah trauma	Terbatas
Gambaran	Polos	Diskus interkalaris	Gelap-terang
Letak	Dinding saluran cerna	Jantung	Ekstremitas





DAFTAR

PUSTAKA

1. Mescher, A.L. (2016) Junquiera's Basic Histology, Text and Atlas 14th edn. New York:McGraw-Hill Education
2. Ereschenko, V.P. 2013. diFiore's Atlas of Histology with Functional Correlations 12th Edition. Philadelphia : Lippincott Williams & Wilkins.

